

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
from google.colab import files

uploaded = files.upload()
```

Choose files

AQI_Data_...Project.ipynb

AQI_Data_Science_Project.ipynb(n/a) - 1834439 bytes, last modified: 12/05/2026 - 100% done

Saving AQI_Data_Science_Project.ipynb to AQI_Data_Science_Project.ipynb

```
import os

print("Files in current folder:")
print(os.listdir())

Files in current folder:
['.config', 'AQI_Data_Science_Project.ipynb', 'sample_data']
```

```
from google.colab import files
import pandas as pd

# Upload file
uploaded = files.upload()

# Automatically read the uploaded CSV
filename = list(uploaded.keys())[0]

df = pd.read_csv(filename)

print("Dataset Loaded Successfully")
print("Filename:", filename)
print("Dataset Shape:", df.shape)

df.head()
```

Choose files

global_urban_aqi_dataset.csv

global_urban_aqi_dataset.csv(text/csv) - 1157862 bytes, last modified: 04/06/2026 - 100% done

Saving global_urban_aqi_dataset.csv to global_urban_aqi_dataset.csv

Dataset Loaded Successfully

Filename: global_urban_aqi_dataset.csv

Dataset Shape: (14184, 13)

	City	Country	Date	AQI	PM2.5	PM10	NO2	S02	CO	O3	Temperature	Humidity	Wind_Speed	
0	Bangkok	Thailand	2019-04-25	145.99	53.12	90.86	22.88	20.63	42.87	29.27	10.3	65.0	5.3	
1	Mexico City	Mexico	2019-07-18	189.57	88.63	152.78	22.43	19.15	60.82	48.39	19.2	46.4	8.7	
2	Cairo	Egypt	2024-05-23	125.34	57.52	107.66	14.76	17.22	31.12	23.56	12.0	65.2	4.4	
3	Shanghai	China	2015-02-19	247.53	93.99	171.95	45.34	42.73	58.79	56.00	23.8	67.4	4.3	
4	Tokyo	Japan	2025-10-16	41.82	20.31	17.83	5.66	5.12	12.72	1.39	19.7	50.4	10.1	

Next steps:

Generate code with df

New interactive sheet

```
def aqi_state(aqi):
    if aqi <= 100:
        return "Low AQI"
    elif aqi <= 200:
        return "Medium AQI"
    else:
        return "High AQI"

df['AQI_State'] = df['AQI'].apply(aqi_state)

df[['AQI', 'AQI_State']].head()
```

	AQI	AQI_State	
0	145.99	Medium AQI	
1	189.57	Medium AQI	
2	125.34	Medium AQI	
3	247.53	High AQI	
4	41.82	Low AQI	

```
state_mapping = {
    "Low AQI":0,
    "Medium AQI":1,
    "High AQI":2
}

df['State'] = df['AQI_State'].map(state_mapping)

df[['AQI_State', 'State']].head()
```

	AQI_State	State	
0	Medium AQI	1	
1	Medium AQI	1	
2	Medium AQI	1	
3	High AQI	2	
4	Low AQI	0	

```
actions = {
    0:"No Restriction",
    1:"Partial Restriction",
    2:"High-Pollution Alert"
}

print(actions)
```

```
{0: 'No Restriction', 1: 'Partial Restriction', 2: 'High-Pollution Alert'}
```

```
reward_table = np.array([
    [10, -2, -5], # Low AQI
    [-6, 10, 2], # Medium AQI
    [-10, -3, 10] # High AQI
])

reward_table

array([[ 10, -2, -5],
       [-6, 10, 2],
       [-10, -3, 10]])
```

```
num_states = 3
num_actions = 3

Q = np.zeros((num_states, num_actions))

print(Q)

[[0. 0. 0.]
 [0. 0. 0.]
 [0. 0. 0.]
```

```
alpha = 0.1
gamma = 0.9
epsilon = 0.1
episodes = 500

rewards_per_episode = []
```

```
for episode in range(episodes):
```

```
total_reward = 0

for state in df['State']:

    if np.random.uniform(0,1) < epsilon:
        action = np.random.randint(num_actions)
    else:
        action = np.argmax(Q[state])

    reward = reward_table[state, action]

    next_state = state

    Q[state, action] = Q[state, action] + alpha * (
        reward + gamma * np.max(Q[next_state]) - Q[state, action]
    )

    total_reward += reward

rewards_per_episode.append(total_reward)

print("Training Completed")
```

Training Completed

```
q_table = pd.DataFrame(
    Q,
    columns=[
        "No Restriction",
        "Partial Restriction",
        "High-Pollution Alert"
    ],
    index=[
        "Low AQI",
        "Medium AQI",
        "High AQI"
    ]
)

q_table
```

	No Restriction	Partial Restriction	High-Pollution Alert	
Low AQI	100.0	88.0	85.0	 
Medium AQI	84.0	100.0	92.0	
High AQI	80.0	87.0	100.0	

Next steps: [Generate code with q_table](#) [New interactive sheet](#)

```
best_actions = np.argmax(Q, axis=1)

states = ["Low AQI", "Medium AQI", "High AQI"]

print("Final Learned Policy\n")

for i, state in enumerate(states):
    print(state, "->", actions[best_actions[i]])
```

Final Learned Policy

Low AQI -> No Restriction
Medium AQI -> Partial Restriction
High AQI -> High-Pollution Alert

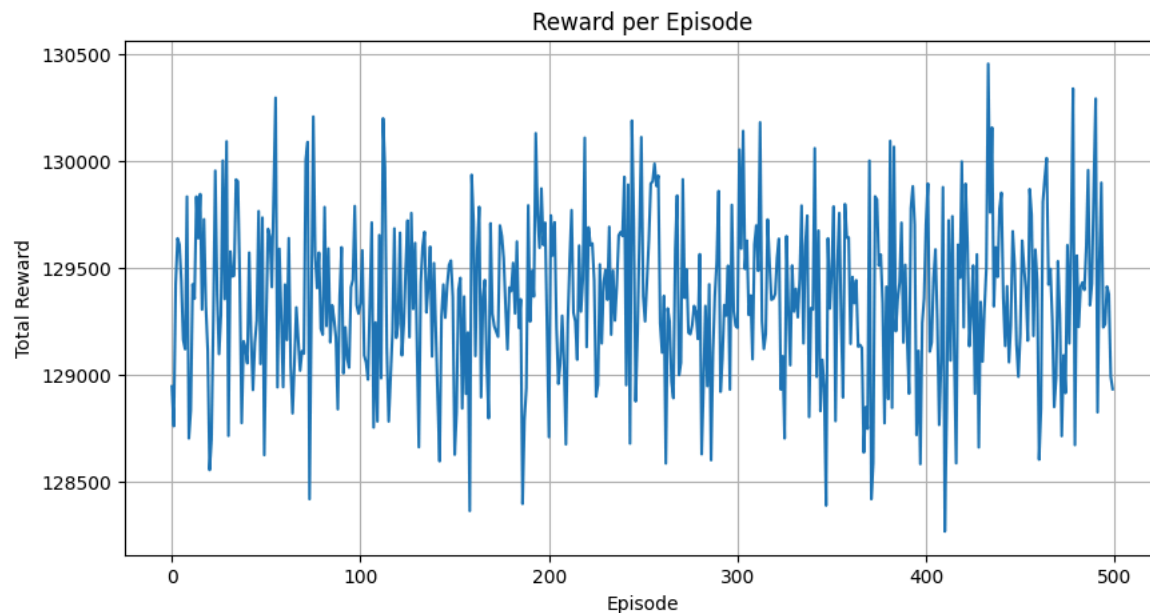
```
average_reward = np.mean(rewards_per_episode)

print("Average Reward After Training:", round(average_reward,2))
```

Average Reward After Training: 129351.52

```
plt.figure(figsize=(10,5))
plt.plot(rewards_per_episode)
plt.title("Reward per Episode")
plt.xlabel("Episode")
```

```
plt.ylabel("Total Reward")
plt.grid(True)
plt.show()
```



```
plt.figure(figsize=(8,4))

plt.imshow(Q, cmap='YlGnBu')

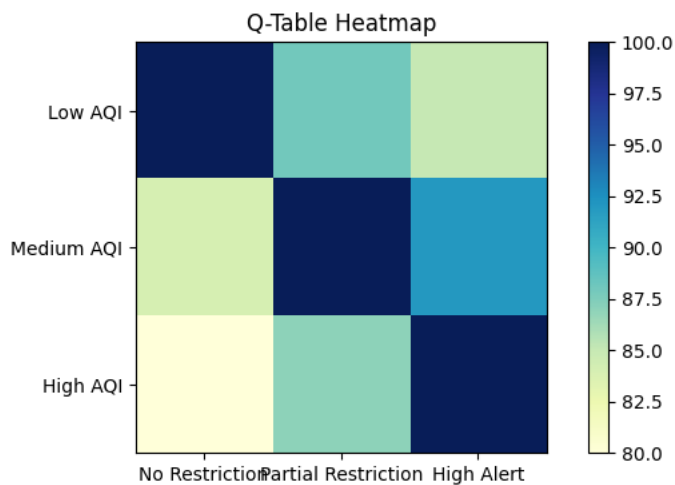
plt.colorbar()

plt.xticks(
    [0,1,2],
    ["No Restriction","Partial Restriction","High Alert"]
)

plt.yticks(
    [0,1,2],
    ["Low AQI","Medium AQI","High AQI"]
)

plt.title("Q-Table Heatmap")

plt.show()
```



```
action_counts = np.bincount(best_actions)

plt.figure(figsize=(6,4))

plt.bar(
    ["No Restriction", "Partial Restriction", "High Alert"],
    action_counts)
```

```
[No Restriction, Partial Restriction, High Alert])  
    action_counts  
)  
  
plt.title("Action Distribution")  
  
plt.ylabel("Count")  
  
plt.show()
```

